Quantifying the Impact of Concurrent Operating Room Turnovers on Hybrid Operating Room Efficiency

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***Abstract*—Mixed linear regression models were applied to quantify how concurrent turnovers impact workflow efficiency in hybrid operating rooms. While equipment reconfigurations and simultaneous turnovers significantly extended overall turnover duration, anesthesia preparation duration was not significantly affected, highlighting precise targets for scheduling optimization.**

***Index Terms*—Concurrent turnovers, equipment reconfiguration, hybrid operating room (OR), mixed linear regression, scheduling optimization**

I. Introduction

Hospitals generate revenue exclusively during incision-to-suture time, making any unproductive operating room (OR) downtime a direct financial loss [1]. Surgical turnover—the window between one patient exiting and the next entering—represents the largest controllable source of idle room time. While standard scheduling models treat ORs as independent units, real-world rooms operate in physical clusters. At LMU Klinikum, each cluster relies on shared resources, including a single attending anesthesiologist per cluster. When neighboring rooms turn over simultaneously, this staffing constraint creates immediate operational bottlenecks.

This project empirically evaluates and verifies key operational observations made by our supervisor regarding these turnover dynamics. Specifically, we test the hypotheses that cluster-level strain inflates turnover time and that mid-day reconfigurations of specialized Siemens Artis Zeego systems in hybrid suites impose measurable downtime penalties.

This project was done as part of TUM course Clinical Applications of Computational Medicine in summer semester 2026.

This project was supervised by Stefanie Grabow (OP-Coordinator LMU Klinikum, MBA Health Management & Specialist in Operating Room Nursing), who facilitated the data provided by the LMU Klinikum.

Beyond the general scientific objective, the project addressed a concrete operational need at LMU Klinikum. In her work on improving the scheduling of the hybrid operating room suites, the project supervisor required a data-driven assessment of whether concurrent turnovers represent a measurable source of inefficiency and delay and, if so, under which operational conditions and to what extent. The analysis was therefore designed to translate this practical concern into quantifiable evidence based on real clinical workflow data.

II. Methodology

All data preprocessing, feature engineering, and statistical modeling—including ordinary least squares and linear mixed-effects regression—were performed following standard methodology for statistical data analysis [2]–[4].

A. Data Acquisition and Preprocessing

After acquiring the clinical dataset from LMU Klinikum [5], data preprocessing was performed to resolve and consolidate parallel timestamps. The raw dataset contained multiple overlapping records for key surgical milestones, specifically categorized into core, documented, and auxiliary timestamps for incision, suture, and patient wake-up events. To verify the integrity of the resulting data and quantify any statistical bias between these recording methods, Wilcoxon signed-rank tests were conducted on timestamp discrepancies. As detailed in Table I, the large sample sizes resulted in highly significant p-values ($p < 0.01$) in the comparisons. However, the actual median biases were clinically negligible, ranging from 0.0 to 2.0 minutes. Following this validation, the parallel timestamps were consolidated sequentially—prioritizing auxiliary records, followed by documented and core entries—to establish a single definitive timeline for each procedure.

TABLE I
WILCOXON SIGNED-RANK TEST RESULTS FOR TIMESTAMP DISCREPANCIES

Event	Comparison	*n*	Med. Bias (min)	*p*-value
Suture				
	Core vs. Doc.	5577	2.0	< 0.001
	Core vs. Aux.	5578	2.0	< 0.001
	Doc. vs. Aux.	5644	0.0	0.008
Incision				
	Core vs. Doc.	5615	2.0	< 0.001
	Core vs. Aux.	5616	2.0	< 0.001
	Doc. vs. Aux.	5657	0.0	< 0.001
Wakeup				
	Core vs. Aux.	4828	0.0	< 0.001

Note: Comparisons with missing columns in the raw data were omitted.

The expected duration of anesthesia preparation was then calculated for each procedure. To achieve this, free-text clinical notes documenting induction methods were parsed using regular expressions. The extracted text was matched with a predefined clinical dictionary comprising base anesthesia techniques, monitoring lines, and regional blocks, each with an associated standard duration in minutes. To ensure accuracy and prevent overestimation, duplicate matches within the same procedure were filtered. The standard durations of all unique identified categories were then summed to establish the total expected anesthesia time for the surgery. A representative subset of these standardized durations, categorized by technique, is provided in Table II.

TABLE II
STANDARD DURATIONS FOR SELECTED ANESTHESIA TECHNIQUES

Category	Procedure / Technique	Duration (min)
Base Anesthesia	General Anesthesia (AA / RSI / IVA)	20
	Double Lumen Tube (DLT)	30
	Spinal / Epidural (SpA / PDA / CSE)	35
	PIZA (Thoracic Epidural + General)	75
Lines & Monitoring	Arterial Line (Art)	10
	Central Venous Catheter (ZVK)	10
	Large Bore Catheter (Schleuse)	10
Regional Blocks	Standard Blocks (e.g., PVB, ISB, SCB)	20
	Transversus Abdominis Plane (TAPB)	15
	Femoral Nerve Block (FemB)	15
	Distal Blocks (e.g., SaphB, FußB)	10

Note: Subset of standardized times used.

B. Defining Concurrent Turnovers (Cluster Strain)

Cluster strain was then measured by identifying concurrent turnover phases within the same operational cluster on a given day. For every turnover, we analyzed parallel activities in neighboring rooms. Initially, the interval from patient wake-up to patient arrival in the holding area was used. After consultation with the project supervisor, this interval was replaced by the time from suture to the subsequent incision. If the turnover window of a neighboring room intersected the active turnover window, it was flagged as a concurrent event. Two primary metrics were computed to capture this strain. Discrete strain level is the absolute count of overlapping turnovers, ranging from 0 (an isolated turnover) to 3 concurrent overlaps. Extreme instances of 4 concurrent turnovers were excluded from the final analysis, as they represented less than 0.1% of the dataset. For the remaining overlaps, their duration was calculated, where the number of overlapping minutes was summed to get a single continuous metric.

C. Phase Segmentation

To isolate specific bottlenecks within the surgical workflow, overall turnover time was segmented into distinct phases following standard definitions from the German Perioperative Procedural Time Glossary [6]. The wake-up phase was defined as the duration from suture completion to the final patient wake-up timestamp. Following the room transition, anesthesia preparation time was isolated as the interval from the documented start of anesthesia to the patient's release for surgery. Finally, to capture the period when the operating room remained unoccupied for cleaning and preparation, OR dark time was calculated as the interval between the wake-up timestamp of the prior patient and the start of anesthesia for the subsequent patient.

D. Tracking Equipment Reconfiguration

In two operating rooms, OR-8 and OR-9, specialized Zeego imaging equipment is utilized. This system can be operated in three primary configurations, designated as Zeego 1 through Zeego 3. To identify equipment reconfigurations, the procedure text descriptions were parsed to determine the active Zeego configuration for each case. Forward-filling was then applied within each operating room's chronological sequence to track state persistence, and any alterations between successive procedures were flagged as discrete reconfiguration events.

E. Statistical Modeling

To evaluate drivers of turnover delays and subinterval durations, linear mixed-effects models were fitted using Restricted Maximum Likelihood (REML). The baseline model structure for turnover time is defined in (1):

$$\text{turnover\_time} \sim \text{cluster\_strain} + \text{priority} + \text{expected\_anesthesia} \tag{1}$$

where priority represents the categorical urgency level ($N1$ being the highest priority through $N5$ lowest priority), cluster_strain represents total overlap duration in minutes, and expected_anesthesia represents expected anesthesia preparation time in minutes. This formulation was adapted for subinterval modeling, with expected_anesthesia omitted for wake-up and dark room models. Models were initially specified with surgical department as a random effect and subsequently evaluated with operational cluster as a random effect. As model fit remained consistent across grouping variables, cluster was selected as the final random grouping factor.

Initial comparisons of Zeego reconfiguration cases used two-sample independent t-tests on unadjusted turnover times between switch and non-switch cases. To control for confounding clinical variables, equipment reconfiguration was evaluated using both linear mixed-effects models and Ordinary Least Squares (OLS) regression. Heteroscedasticity-robust standard errors (HC3) were incorporated into the OLS specifications to account for non-constant variance in turnover durations. Finally, to evaluate the relationship under standard operational conditions, log-linear OLS specifications restricted to standard turnover durations (≤ 240 minutes) were estimated.

III. RESULTS

A. Baseline Operational Metrics

The final dataset comprised 3,629 retained turnover events across the operational clusters, representing 99.0% of all logged turnovers after filtering extreme strain outliers. Baseline behavior varied substantially across the two clusters, as detailed in Table III.

Cluster 0 ($n = 1,495$) demonstrated a median turnover duration of 111.0 minutes, whereas Cluster 1 ($n = 2,134$) exhibited longer transitions with a median of 138.0 minutes. Evaluating the underlying subintervals reveals that while anesthesia preparation and wake-up times remained relatively consistent across locations (medians of 21.0–23.0 minutes and 16.0–18.0 minutes, respectively), unoccupied OR duration (dark time) accounted for the primary divergence between the clusters, recording medians of 35.0 minutes in Cluster 0 versus 93.0 minutes in Cluster 1.

TABLE III
BASELINE OPERATIONAL METRICS ACROSS CLUSTERS

Metric	*n*	Med. (min)	Mean ± SD (min)	P95 (min)
Turnover				
Cluster 0	1495	111.0	177.4 ± 250.2	575.2
Cluster 1	2134	138.0	229.8 ± 250.2	625.3
Anesthesia				
Cluster 0	1379	21.0	25.2 ± 25.0	64.1
Cluster 1	1995	23.0	28.4 ± 28.0	65.0
Wake-up				
Cluster 0	1310	18.0	21.6 ± 18.0	53.0
Cluster 1	1918	16.0	20.6 ± 15.0	51.0
Dark room				
Cluster 0	1130	35.0	109.2 ± 273.0	518.4
Cluster 1	1571	93.0	179.8 ± 273.0	564.5

Note: P95 denotes the 95th percentile duration.

B. The Impact of Cluster Strain on Turnovers

Cluster strain demonstrated a statistically significant effect on overall turnover durations. The linear mixed-effects model controlling for surgical priority, expected anesthesia complexity, and random cluster effects ($n = 3,063$) showed that total overlap duration positively predicted turnover times. As detailed in Table IV, every additional minute of concurrent turnover overlap significantly prolonged total turnover time ($\beta = 0.334$, $SE = 0.033$, $p < 0.001$). Expected anesthesia preparation time was also a statistically significant positive predictor ($\beta = 0.160$, $SE = 0.058$, $p = 0.006$).

In contrast, surgical priority levels ($N1$ through $N5$) did not demonstrate statistically significant effects on turnover delays when compared to emergency baselines ($p > 0.10$ across all priority categories).

TABLE IV
MIXED LINEAR REGRESSION MODEL FOR TOTAL TURNOVER TIME

Variable	Coef. (β)	Std. Err.	*z*	***p*-value**
Intercept	217.85	63.79	3.415	0.001
Cluster Strain (min)	0.334	0.033	10.195	< 0.001
Exp. Anesthesia (min)	0.160	0.058	2.755	0.006
Priority N1	33.16	65.17	0.509	0.611
Priority N2	4.46	63.24	0.071	0.944
Priority N3	5.89	63.03	0.093	0.926
Priority N4	-73.02	63.27	-1.154	0.248
Priority N5	-96.37	62.84	-1.534	0.125

Model Fit: $n = 3,063$, Log-Likelihood: -21532, Scale: 82898, Group Var: 258.99.

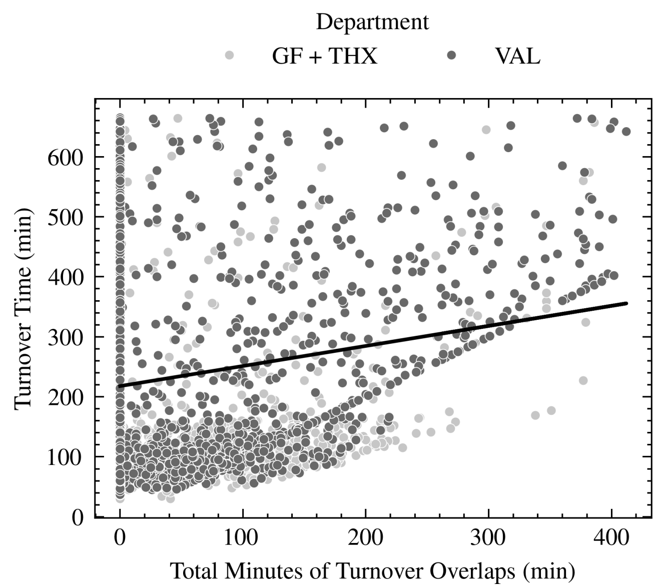

Fig. 1. Overall turnover duration plotted against total cluster turnover overlap minutes, showing a significant positive trend across surgical departments.

C. Phase-Specific Vulnerabilities

Evaluating the individual subintervals revealed where operational strain accumulates during concurrent transitions. As summarized in Table V, hallway cluster strain impacts workflow phases differently across the turnover pipeline:

The anesthesia preparation phase ($n = 3,014$) showed a statistically significant association with cluster strain ($p < 0.001$), but the effect size was clinically negligible ($\beta = 0.018$ minutes per minute of strain). Instead, anesthesia preparation duration was primarily driven by procedure-defined clinical complexity ($\beta = 0.214$, $p < 0.001$).

The patient wake-up phase ($n = 3,212$) showed no measurable association with cluster strain ($\beta = 0.006$, $p = 0.117$).

The unoccupied OR duration (dark time, $n = 3,085$) absorbed the vast majority of cluster congestion. Each additional minute of concurrent turnover overlap generated 0.561 minutes of OR dark time ($\beta = 0.561$, $SE = 0.072$, $p < 0.001$).

TABLE V
COMPARISON OF MIXED LINEAR MODELS ACROSS WORKFLOW PHASES

Phase / Variable	Coef. (β)	Std. Err.	*p*-value	Model Fit
Anesthesia Prep				$n = 3014$
Intercept	5.640	6.243	0.366	L-L: -12168
Cluster Strain (min)	0.018	0.003	< 0.001	Scale: 187.79
Expected Anesthesia	0.214	0.006	< 0.001	
Wake-up Phase				$n = 3212$
Intercept	23.204	3.957	< 0.001	L-L: -13684
Cluster Strain (min)	0.006	0.004	0.117	Scale: 294.19
Dark / Empty Time				$n = 3085$
Intercept	192.303	85.139	0.024	L-L: -22746
Cluster Strain (min)	0.561	0.072	< 0.001	Scale: 151098

L-L denotes Log-likelihood
Note: All models control for surgical priority and random department effects.

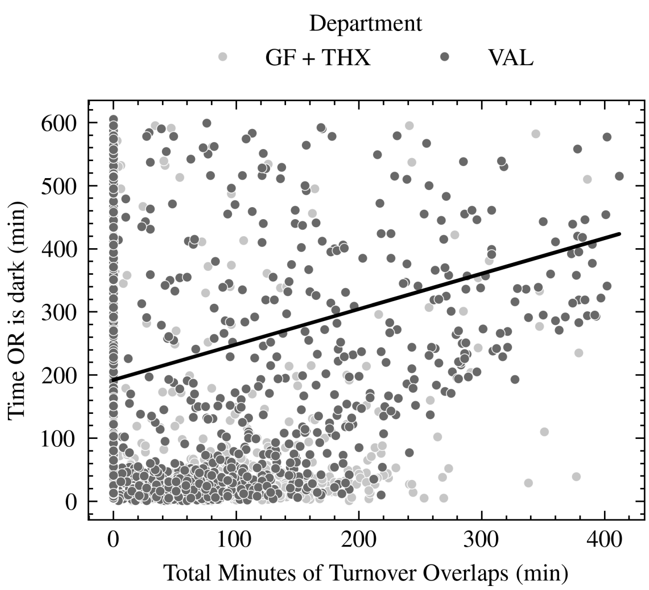

Fig. 2. Linear trend demonstrating the positive relationship between total cluster turnover overlap duration and OR dark time across departments.

D. Equipment Reconfiguration Penalties

In hybrid operating rooms equipped with specialized Zeego systems (OP2-08 and OP2-09), unadjusted turnover times averaged 172.4 minutes for non-switch procedures versus 193.1 minutes when an equipment reconfiguration occurred, representing a raw operational difference of 20.8 minutes ($p = 0.0630$).

While unconstrained regression models failed to reach statistical significance across all turnover ranges ($\beta = 0.0677$, $p = 0.121$), the optimized log-linear OLS specification restricted to standard operating conditions (≤ 240 minutes, $n = 757$) revealed a clear, statistically significant reconfiguration penalty. As detailed in Table VI, executing a Zeego configuration switch added an average of 11.486 minutes to standard turnover duration ($\beta = 11.486$, $SE = 5.993$, $p = 0.032$). Under typical workflow constraints, this equipment transition

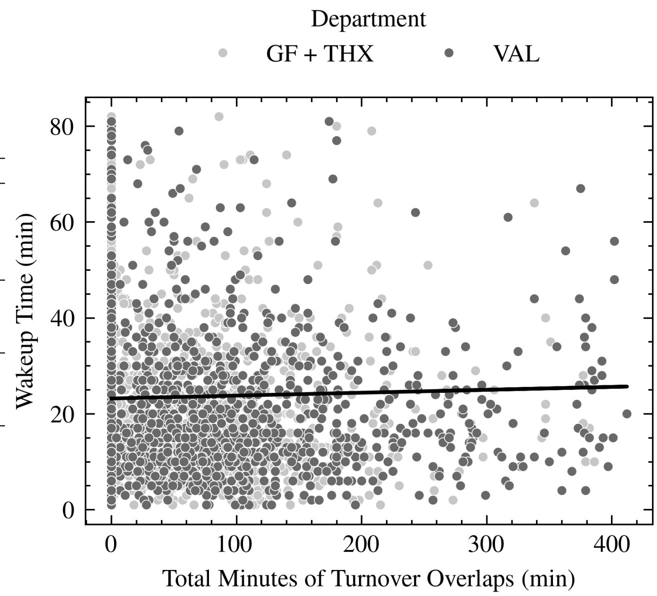

Fig. 3. Scatter plot and fitted trend line illustrating the resilience of patient wake-up duration against increasing cluster strain minutes.

represents an approximate +7.0% relative increase in baseline room downtime.

TABLE VI
HYPER-OPTIMIZED OLS REGRESSION MODEL FOR ZEEGO RECONFIGURATION (≤ 240 MIN)

Variable	Coef. (β)	Std. Err.	*t*	*p*-value
Intercept	73.181	12.014	6.091	< 0.001
Zeego Switch (*binary*)	11.486	5.993	2.138	0.032
Cluster Strain (min)	0.177	0.028	6.223	< 0.001
Exp. Anesthesia (min)	0.184	0.076	2.423	0.016
Priority N1	-15.421	14.881	-1.036	0.300
Priority N2	-8.914	12.302	-0.725	0.469
Priority N3	-1.205	12.185	-0.099	0.921
Priority N4	-21.408	13.011	-1.645	0.100
Priority N5	-32.115	12.540	-2.561	0.011

Model Fit: $n = 757$, $R^2 = 0.202$, $R^2_{\text{adj}} = 0.194$, $F(8, 748) = 23.68$ ($p < 0.001$).
Note: Heteroscedasticity-robust standard errors (HC3) applied.

IV. DISCUSSION

A. Operational Drivers of Turnover Delays

The findings demonstrate that turnover delays are primarily driven by the overutilization of shared support staff across an OR cluster, rather than direct patient-care procedures within individual operating rooms. While clinical tasks requiring direct patient interaction—such as anesthesia induction and postoperative emergence—remain consistent during high-demand periods, idle room time (dark time) absorbs nearly all cluster-level congestion. This indicates that bottlenecks in shared logistical support, including room turnover teams, patient transport personnel, and equipment preparation staff, manifest as unproductive room delays rather than disruptions to direct clinical care.

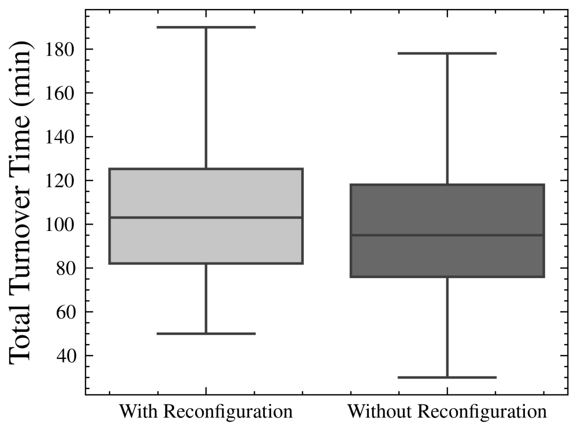

Fig. 4. Comparison of total turnover times in hybrid OR suites with and without Zeego equipment reconfiguration.

Additionally, non-standard room setups introduce consistent operational friction. In hybrid operating suites, switching between specialized Zeego configurations imposes a statistically significant penalty of approximately 11.5 minutes per transition under standard operating conditions. While modest for an isolated procedure, these resets accumulate across multi-room surgical schedules, further straining support personnel during peak operational windows.

B. Limitations

This project has several limitations. First, while the optimized model for Zeego reconfiguration yielded statistically significant estimates, the modest coefficient of determination ($R^2 = 0.202$, $R^2_{\text{adj}} = 0.194$) indicates that a substantial portion of the variance in surgical turnover times remains unexplained by routine scheduling metrics. In complex clinical environments, uncaptured stochastic factors—such as patient physiological variation, unmeasured intraoperative complications, variable team composition, and real-time staffing shortages—introduce significant baseline noise. Incorporating advanced data normalization techniques in future work, such as standardizing turnover metrics relative to specific surgical specialties, case complexities, and overall cluster strain, could better account for baseline variance and further elevate the predictive power of the models.

Second, while the dataset provided high-resolution timestamps for surgical milestones, it lacked detailed logging of support staff assignments. This constraint prevented a clear distinction between the overutilization of support staff and the underutilization of an operating room due to external logistical delays. Third, evaluating equipment reconfigurations required filtering extreme outlier delays (> 240 minutes) to model standard operational behavior; future investigations should specifically explore the root causes driving these severe, non-standard disruptions. Finally, as a single-center study conducted across two operational clusters at LMU Klinikum, local spatial layouts and institutional workflows may influence absolute durations, warranting cross-institutional validation.

V. Conclusion

This project evaluated the operational dynamics of surgical turnovers across multi-room operating suites at LMU Klinikum, identifying concurrent cluster strain and equipment reconfigurations as key drivers of turnover delays. Through phase segmentation and mixed linear modeling, hallway congestion was shown to primarily inflate idle operating room dark time while leaving direct patient-care subintervals—such as anesthesia preparation and wake-up phases—largely resilient. Additionally, specialized equipment reconfigurations in hybrid operating rooms were quantified as imparting an 11.5-minute penalty per transition under standard operating conditions (≤ 240 minutes).

These findings indicate that turnover bottlenecks stem predominantly from the overutilization of shared cluster support personnel rather than procedures occurring inside the room. The findings suggest that managers may reduce idle room time by adopting cluster-aware scheduling and minimizing unnecessary equipment reconfigurations.

Beyond quantifying these effects, the results provide the project supervisor with an evidence-based foundation for communicating the limitations of the current scheduling approach to physicians and other operational staff. By demonstrating statistically that concurrent turnovers are associated with measurable delays and by identifying the workflow phases most affected, the findings can support concrete change initiatives, particularly the more deliberate staggering of expected procedure end times and turnovers. In this way, the analysis can contribute to moving hybrid OR scheduling from experience-based coordination toward a more transparent and data-driven approach.

Acknowledgment

The authors express their sincere gratitude to project supervisor Stefanie Grabow for her valuable guidance, for facilitating access to the source data, and for organizing a tour of LMU Klinikum, which was essential for gaining a deeper understanding of the operational environment.

The authors would also like to thank Prof. Dr. Martin Daumer for organizing and teaching the course Clinical Applications of Computational Medicine and for providing the opportunity to carry out this project.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this report, the authors used Gemini and Writefull for language refinement, rephrasing, and manuscript polishing. After using these tools, the authors reviewed and edited the content as necessary and take full responsibility for the final content of the publication.

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